

LunaLink Simulation - Project X

Mission: fixed 500 x 36,000 km, 63.4 deg HEO with Ottobrunn ground station.

Scope: all four subsystems are simulated: EPS, TCS, ADCS, and TT&C.

Tool: Python 3.13, headless evidence generator, Streamlit dashboard, 74 tests.

High-fidelity physics: luni-solar + J3 orbit with the 63.4 deg critical-

inclination frozen-apsides demonstration; Van Allen belt dose and 5-year solar-

array degradation; closed-loop PD sun-pointing; ITU-R P.618/676 rain+gas attenuation,

CCSDS coding gain and Doppler.

Independent validation: the orbit is cross-checked against Orekit 13.1 (8x8 gravity +

luni-solar) and NASA SPICE DE440 - period exact, J2-only within 38 km over 36 h,

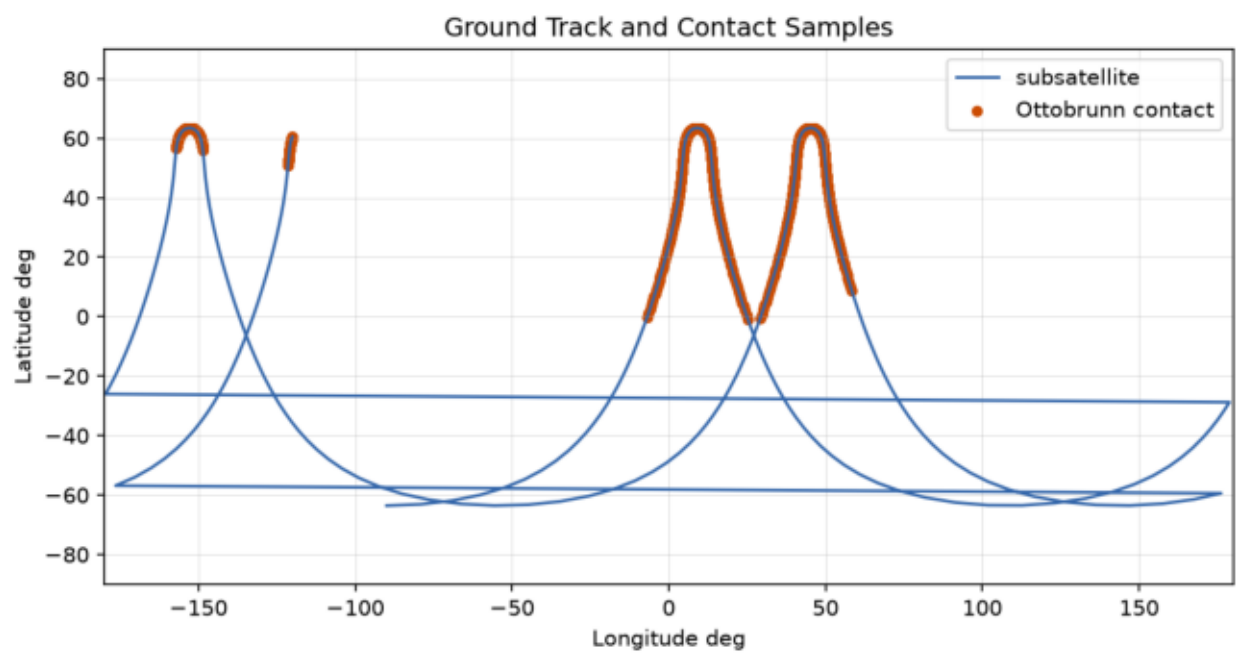
argument-of-perigee drift 0.004 deg/day at 63.4 deg vs 0.29 at 45 deg.

Claim boundary: preliminary engineering simulator, not flight-qualified.

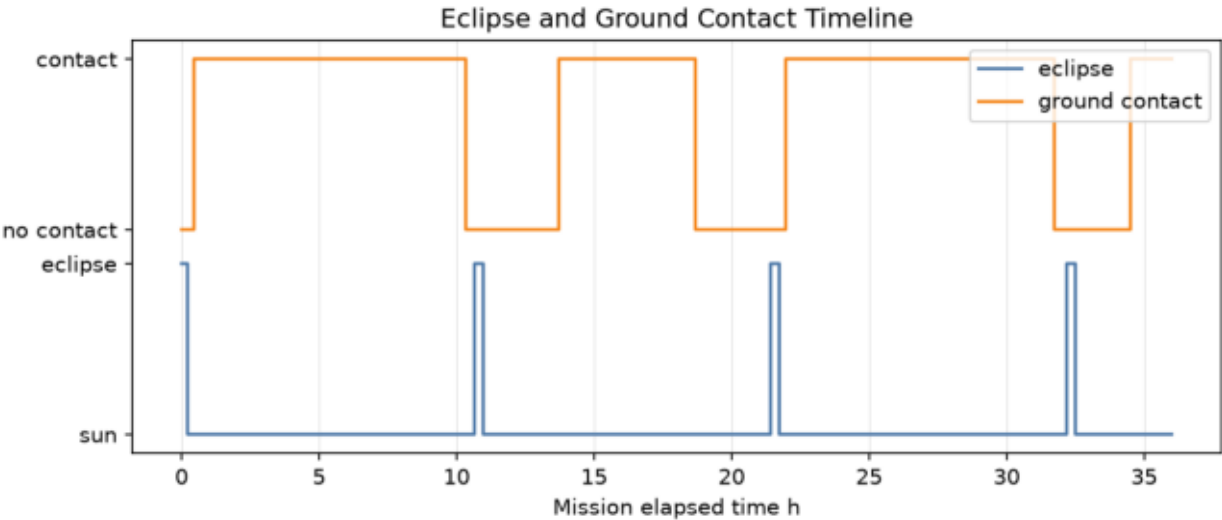
Validation Summary

name	status	value
fixed_altitude_orbit_period_h	pass	10.684539794233128
simulation_duration_h	pass	36.0
minimum_altitude_km	pass	500.0000000000009
maximum_altitude_km	pass	36229.9574504823
orbit_critical_inclination_deg	pass	63.4
orbit_frozen_apsides_argp_rate_deg	pass	0.0004654701051559739
minimum_contact_elevation_rad	pass	0.08762296873739589
eclipse_samples	pass	70
eps_minimum_state_of_charge	pass	0.7538966049382716
eps_unserved_energy_j	pass	0.0
eps_array_eol_power_w	pass	1927.0126800000003
eps_peak_load_w	pass	1200.0
eps_average_load_w	pass	1098.3796296296296
thermal_component_limit_flag	pass	False
thermal_worst_operating_margin_k	pass	7.632638911303559
adcs_final_angular_speed_deg_s	pass	0.025901690354961412
adcs_wheel_saturated	pass	False
ttc_xband_min_margin_db	pass	5.12739305815262
ttc_uhf_min_margin_db	pass	5.4934794046042725
ttc_xband_data_rate_bps	pass	100000000.0
adcs_sun_pointing_settled_error_deg	pass	0.011217117519769608
adcs_dipole_vs_igrf_max_rel_diff	pass	0.3859594901502412
ttc_xband_atmos_loss_5deg_db	pass	2.5741244290145273
ttc_ccsds_coding_gain_db	pass	8.0
radiation_annual_dose_krad	pass	15.254112003494381
radiation_array_remaining_power_5y	pass	0.9486442431951079

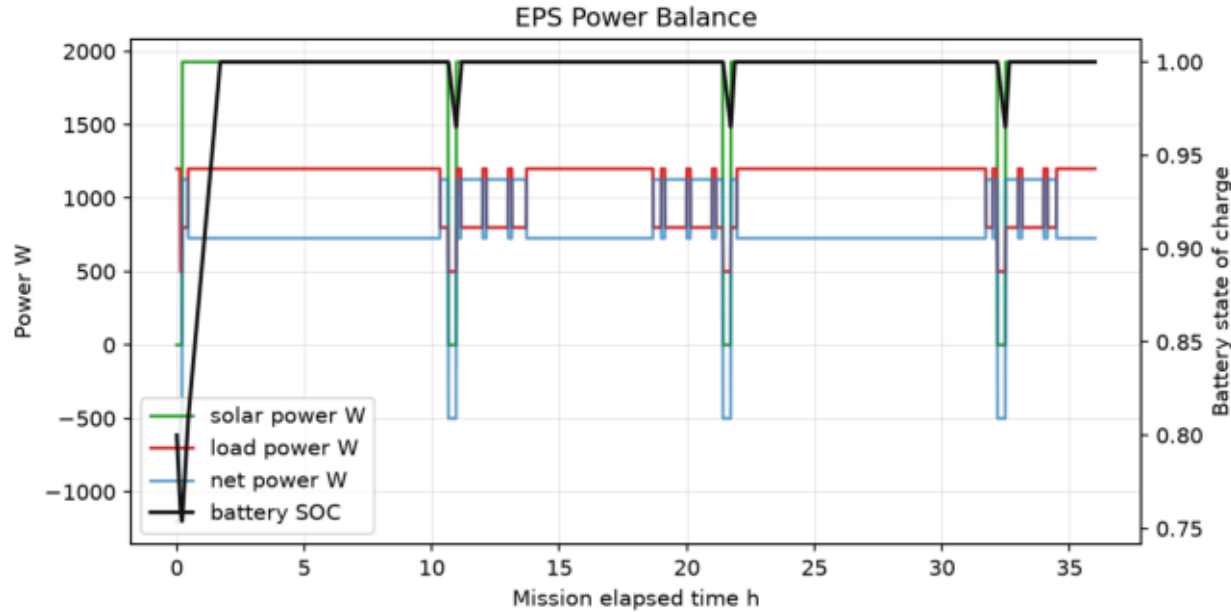
Orbit And Contact Evidence



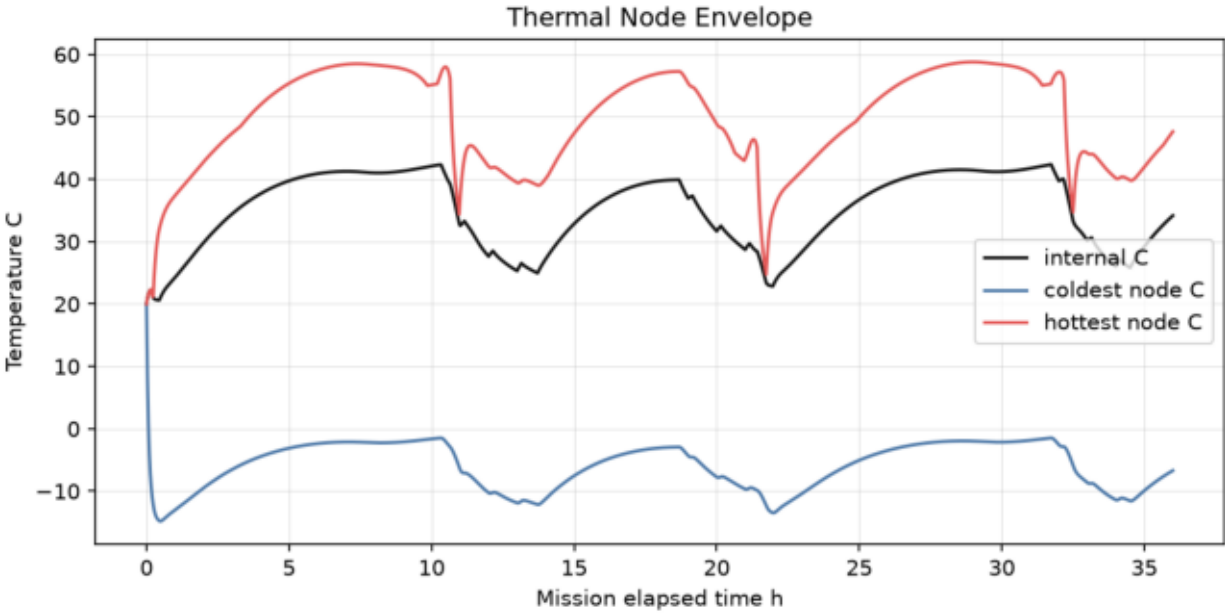
Eclipse And Contact Timeline



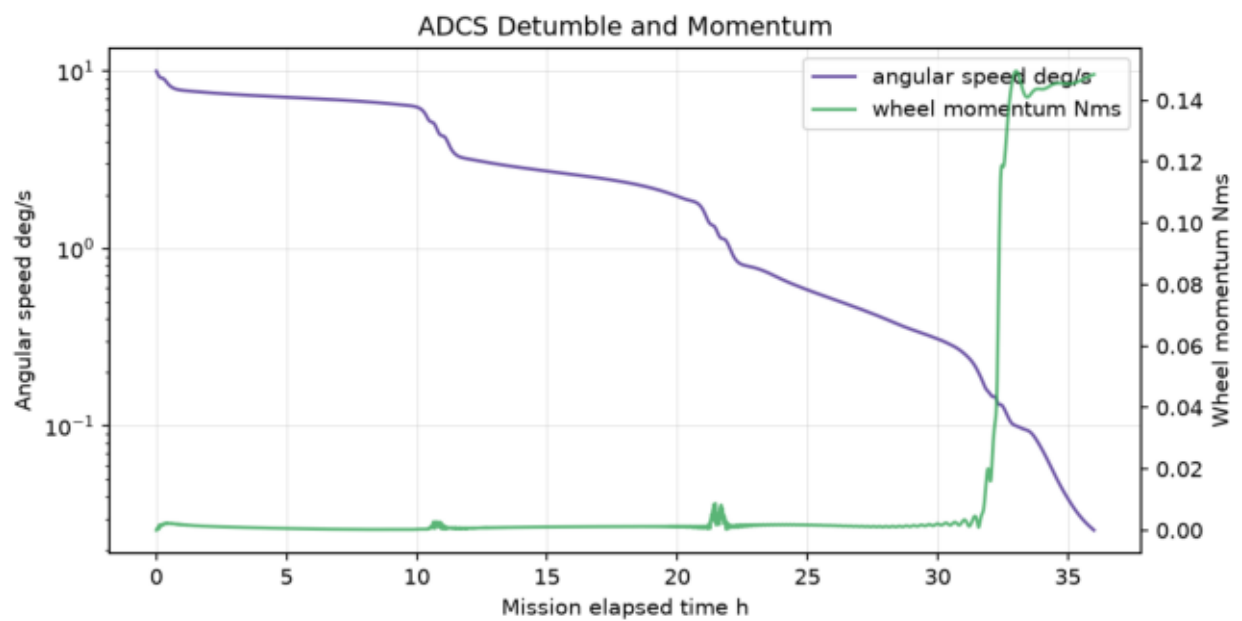
EPS Power And Battery SOC



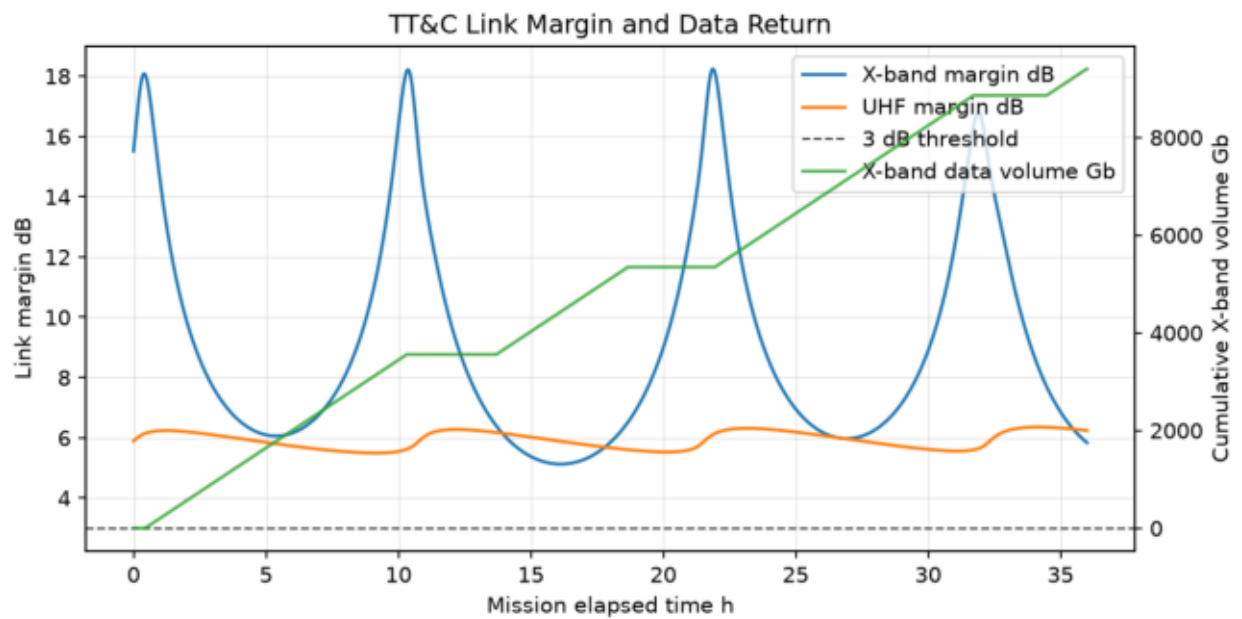
Thermal Six-Face/Internal Response



ADCS Detumble And Momentum



TT&C Range, Margin, And Data



Traceability Snapshot

requirement_id	formula_id	reference	implementation_path	test_path	validation_metrics	evidence_artifact
REQ-MIS-001	F-ORB-001,F-OP	Vallado; NASA S	code/lunalink/or	code/tests/test_	fixed_altitude_o	validation_metrics.csv; [...]
REQ-MIS-003	F-GEO-001,F-FR	Vallado; WGS84	code/lunalink/fr	code/tests/test_	minimum_conta	mission_timeseries.csv; ttc_time
REQ-EPS-001	F-EPS-001,F-EP	NASA SmallSat	code/lunalink/e	code/tests/test_	eps_minimum_s	eps_timeseries.csv; validation_r
REQ-TCS-001,R	F-TCS-001,F-TCS	Gilmore Spacec	code/lunalink/th	code/tests/test_	thermal_compo	thermal_timeseries.csv; [...]
REQ-ADCS-001	F-ADCS-001,F-A	Wertz SMAD; NA	code/lunalink/ad	code/tests/test_	adcs_final_angu	adcs_timeseries.csv; [...]
REQ-TTC-001,R	F-TTC-001,F-TTC	CCSDS RF recor	code/lunalink/tt	code/tests/test_	ttc_xband_min	ttc_timeseries.csv; validation_m
REQ-TRADE-001	F-TRADE-001	Design of exper	code/lunalink/tr	code/tests/test_	reproducible fix	lhs_samples.csv; monte_carlo_s
REQ-XCHK-001	F-ORB-001,F-GE	GMAT/Orekit ind	code/lunalink/ex	code/tests/test_	external correla	[...]

Assumptions, AI Use, And References

All mission parameters are fixed by the Project X brief. SI units are used internally.

The tool states assumptions explicitly in qualification/assumptions_register.csv.

AI was used for planning, implementation assistance, review, and documentation; outputs were checked with tests, static analysis, and engineering sanity checks.

Key references: NASA Systems Engineering Handbook, NASA-STD-7009B, NASA-STD-8739.8B, ECSS-E-ST-10C, ECSS-Q-ST-80C, Vallado, Wertz, Gilmore, Markley and Crassidis, Sidi, CCSDS, JPL DESCANSO, GMAT, Orekit, Basilisk.

EPS minimum SOC: 0.754.

Thermal worst operating margin K: 7.633.

ADCS final angular speed deg/s: 0.0259.

TT&C X-band minimum margin dB: 5.127.

High-fidelity results: analytic apsidal drift 0.0005 deg/day (Orekit 0.004); belt dose 15.3 krad(Si)/yr with 95% 5-yr array power; closed-loop sun-pointing settled error 0.011 deg; ITU-R X-band 5 deg loss 2.57 dB, CCSDS coding gain 8.0 dB.

Open certification items: external GMAT/Orekit numerical correlation, independent review sign-off, hardware-datasheet replacement, and authority acceptance.